



COURSE SYLLABUS						
Cours e Code	ITEC 102	Course Title	Basic Computer Hardware Servicing		Course Credit	3 UNITS
Course Description		This course is designed to develop knowledge, skills, and attitudes of a Computer Service Technician in accordance with industry standards. It covers basic and common competencies such as installing, maintaining, configuring, and diagnosing computer systems. The course is delivered through a combination of strategically designed face-to-face and online (synchronous and asynchronous) sessions.				
Pre-Requisites		None	Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



INSTITUTIONAL LEARNING OUTCOMES (ILO)

As a polytechnic state university, PUP shall develop its students to possess:

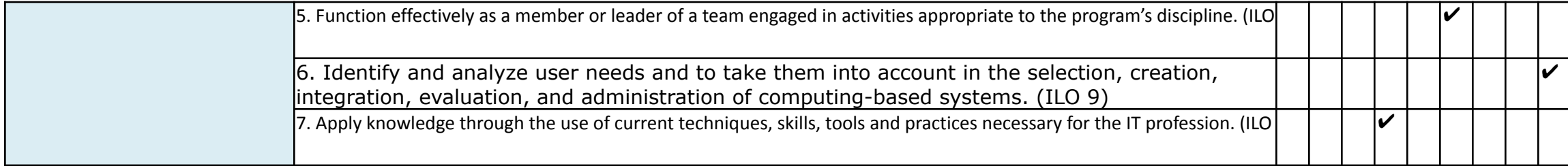
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11.Produce technologies for commercialization of livelihood improvement.

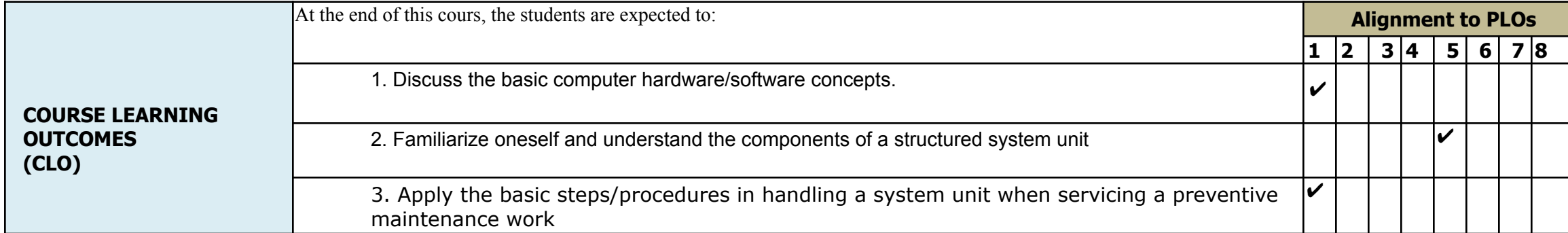


PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
	1. Analyze well-defined computing problem and to apply principles of computing and other relevant disciplines to identify solutions. (ILO 1)	✓								
	2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program’s discipline. (ILO 5)					✓				
	3. Communicate effectively in a variety of professional contexts. (ILO 2)		✓							
	4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles. (ILO7)							✓		



THE COUNTRY'S 1st POLYTECHNICU





THE COUNTRY'S 1st POLYTECHNICU





OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1	1. Understand all the policies and in-house classroom management of the professor. 2. Make students at ease and have an eagerness to learn new knowledge in the world of computers. 3. Acquiring the relation of the subject to their course and ultimately give importance to each topic to be discussed in the succeeding meetings.	- Discussion of Vision, Mission, and Core Values of the Polytechnic University of the Philippines - Discussion of Outcomes-based Teaching and Learning Plan, computation of grades, classroom management, and other requirements of the course - Overview of the topic Basic Computer Hardware Servicing		Synchronous: Teacher activity: Explain the parts of the course syllabus	1) Course evaluation Procedure a Grading system 2) assignments, exams, and other requirements with due dates.		X			
2	1. Describe the importance of system unit service. 2. Show interest and admiration in forging a computer-based society	Basic introduction to computer hardware servicing as in hand-to hand troubleshooting techniques.								



		Basic computing concepts: computer, hardware, Software, and program.								
3-6	- Identify the basic components of a desktop computer and its peripherals as in use at home and in the business sector. - CPU TIMELINE: INTEL and AMD. - Differentiate and identify the function of motherboard and layout, (form factor) arrangement of main parts that differ to another brand. - Identify and familiarize the FOUNDERS/IN VESTORS	- The hardware (motherboard) main parts: Motherboard and its main component parts and functions - The CPU and its socket - The memory module and its slots The AGP/PCI-E card and its port The chipsets - The IDE/Sata controllers - The IDE/Sata controllers - The motherboard power(20+4pin) - The CPU power(4pin), CMOS(Bios) chipset and battery, Parallel/printer port, VGA port, P/S 2 Keyboard and Mouse port, audio ports, Sata ports and LAN port (Ethernet) HDMI port, USB ports.								
7	Identify the basic components of a system unit and its peripherals, Processor and its timeline, Hard Drive: IDE And SATA, Memory module: SIMM/DIMM (DDR, DDR2,	HARDWARE: Basic parts that assemble a system unit, Input / output devices ACRONYMS AND TERMS - Display - Video Memory								



	DDR3, DDR4), Video/Display/G raphics card, Input/output as defined in relationship with computer, 2. Display adapters and monitors. • Describe the functionality of a display adapter. • Outline current and legacy display adapter standards. • Understand how video memory affects display properties. • Outline recent developments in display adapter technology. • Understand monitor specifications. • Troubleshoot basic adapter/display subsystem • Recognize the characteristics of different printer types. • Appreciate the use of printer control languages. • Understand the features of different paper feed mechanisms. • Assess the consumables required for different printer types.	• Monitors • Troubleshooting display/monitor problems • Printer types • Dot matrix • Ink jet or bubble jet • Laser printers • Printer consumables • Printer paper • Printer troubleshooting • Printer command languages • Troubleshooting dot matrix printers • Troubleshooting laser printers • Routine maintenance of laser printers • Troubleshooting ink jet printers								
--	---	---	--	--	--	--	--	--	--	--



9										
10-12	1. Attaching different cables on the rear- panel ports of a system unit	Hands-on practice on the assembly/disassembly of the rear panel ports wiring. q								
13-17	1. Simple safeguarding to prolong the life span of the unit that can be applied to any personal or business office equipment. 2. Can be a part-time job: Applicable to computer shops and rentals. 3. Small computer-based office.	Preventive Maintenance: Maintaining the system unit in good, working condition.								
18	FINAL EXAMINATION									



**POLYTECHNIC UNIVERSITY OF THE
PHILIPPINES**
**OFFICE OF THE VICE PRESIDENT FOR
CAMPUSES**
TAGUIG CAMPUS
**Bachelor of Science in Information Technology
(BSIT)**



General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

THE COUNTRY'S 1st POLYTECHNICU





**POLYTECHNIC UNIVERSITY OF THE
PHILIPPINES**
**OFFICE OF THE VICE PRESIDENT FOR
CAMPUSES**
TAGUIG CAMPUS
**Bachelor of Science in Information Technology
(BSIT)**



General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

THE COUNTRY'S 1st POLYTECHNICU





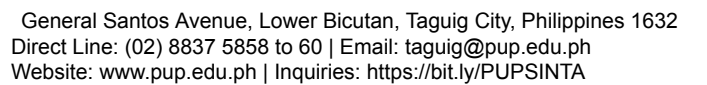
WEBSITE:

THE COUNTRY'S 1st POLYTECHNICU





CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



THE COUNTRY'S 1st POLYTECHNICU





The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.

We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.

Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.

Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic

dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.

Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.

Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.

Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will



**POLYTECHNIC UNIVERSITY OF THE
PHILIPPINES**
**OFFICE OF THE VICE PRESIDENT FOR
CAMPUSES**
TAGUIG CAMPUS
**Bachelor of Science in Information Technology
(BSIT)**



	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
--	---

General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: https://bit.ly/PUPSINTA

THE COUNTRY’S 1st POLYTECHNICU



2025
7th in the Philippines
92nd in Southeast Asia
541-560 in Asia



WURI
The WORLD
UNIVERSITY
RANKINGS
for INNOVATION

Rank 21
GENERATIVE AI

Rank 39
STUDENT SUPPORT

Rank 50
SYMBOL/



QS STARS

★★★★★
Employability

★★★★★
Teaching

★★★★★
Social Responsibility

★★★★★
Online Learning

★★★★★
Academic Development

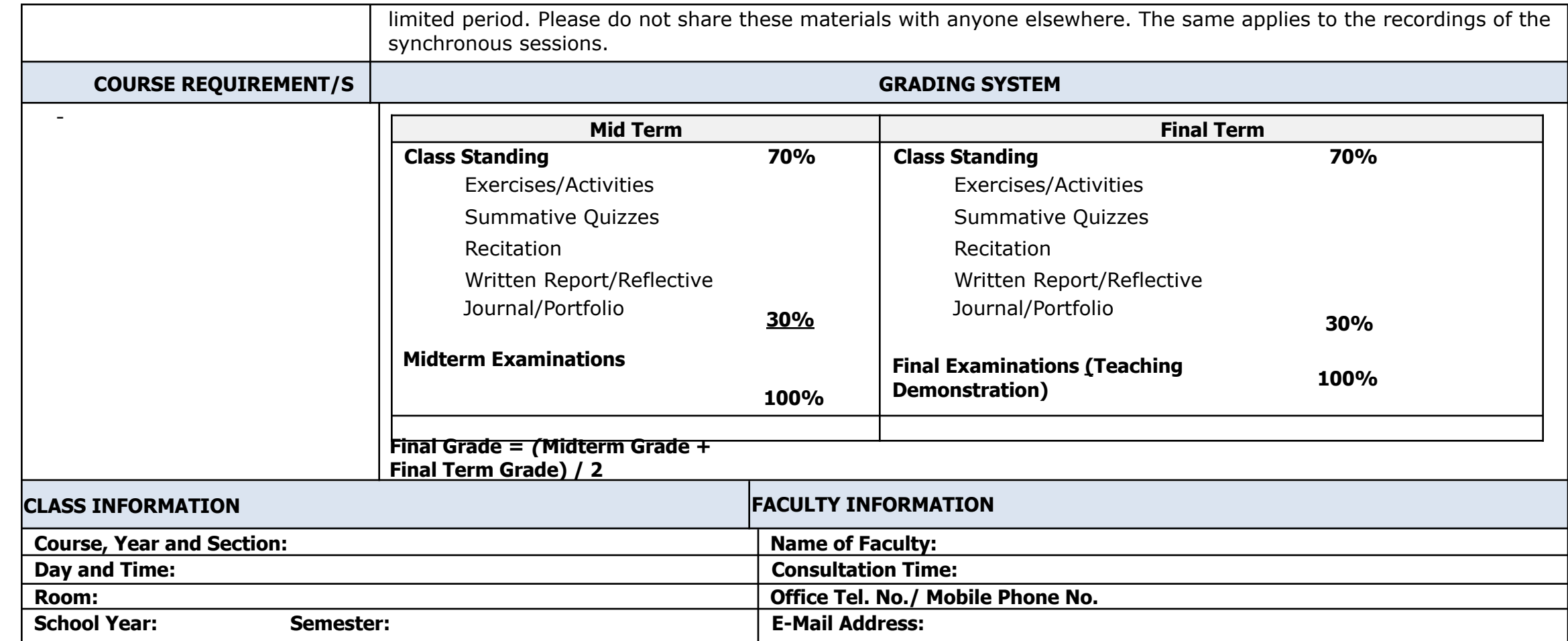
★★★★★
Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

★★★★★
Inclusiveness

★★★★★
Internationalization



**Times Higher Education
Impact Rankings 2024**
DEBUTS AT 1501+



General Santos Avenue, Lower Bicutan, Taguig City, Philippines 1632
Direct Line: (02) 8837 5858 to 60 | Email: taguig@pup.edu.ph
Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

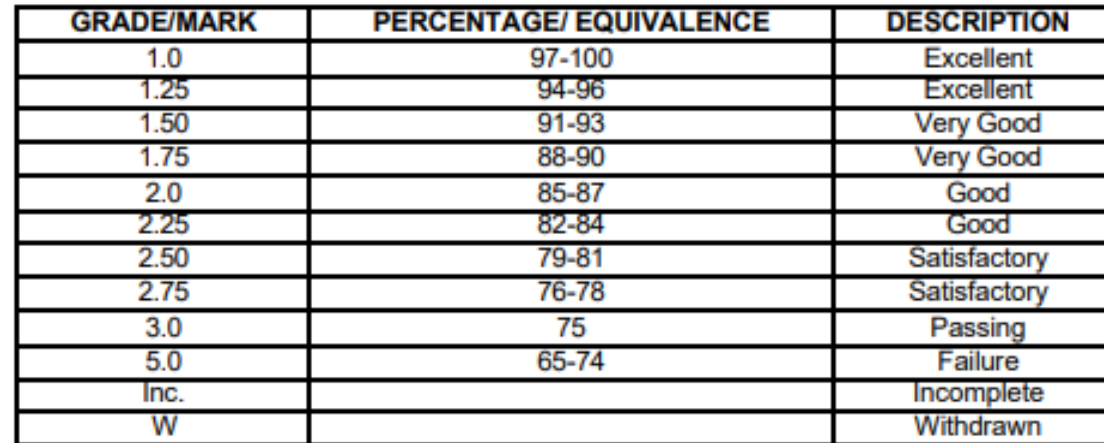
THE COUNTRY'S 1st POLYTECHNICU





Computer Hardware Servicing				
	Excellent 20 pts	Good 17.5 pts	Fair 15 pts	Unsatisfactory 13 pts
Identify On-Board Computer Student's ability to identify on-board computers and properly name computer.	Excellent Demonstrated mastery of task. Student identified and labeled all computers accurately	Good Student identified and accurately named most of the on-board computers. Student missed up to two computers.	Fair Student identified some of the on-board computers. Student missed up to four computers and did not name the computers accurately.	Unsatisfactory Student is unable to identify the amount of on-board computers or missed more than four computers.
Retrieve Trouble Codes Student's ability to navigate the scan tool to retrieve codes.	Excellent Demonstrated mastery of task. No assistance needed to navigate scan tool and retrieve trouble codes.	Good Student is able navigate scan tool and retrieve trouble codes with minimal assistance. May have missed a key point.	Fair Student needed moderate assistance to navigate scan tool and retrieve trouble codes. Student needs to practice more.	Unsatisfactory Student made no effort to retrieve trouble codes. Student displayed no exposure.
Follow Directions of task Student performed task in the order of the directions of the task.	Excellent Demonstrated mastery of task. Student performed task exactly according to directions in task sheet.	Good Student completed task sheet questions, but did not follow the directions exactly.	Fair Student completed task sheet questions, but skipped the majority of the directions of the task.	Unsatisfactory Student did not follow directions of the task. Student filled out task sheet after completing oil change task.
Follows Shop Safety Rules Student follows safety rules of shop. i.e. wears safety glasses, operates equipment safely.	Excellent Student follows all safety rules in the shop and works in a safe manner.	Good Student works safely, needs reminders to keep safety glasses on.	Fair More frequent reminders to keep safety glasses on. Does not always work in a safe manner.	Unsatisfactory Student cannot work in the shop due to unsafe work habits.
Clean -Up Student cleans work area. i.e. put away tools and wipe down work area.	Excellent Student cleaned entire work area with no assistance.	Good Student cleaned work area with prompting.	Fair Student attempted to clean work area, but needed to be prompted.	Unsatisfactory Student did not attempt put tools away or clean work area.





Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
ASSOC. PROF. JOSEPHINE M. DELA ISLA INSTR. ONARD JOHN M. DELA ISLA Faculty/Committee/Date	MS. DIANNE S. SEGURORA Head of Academic Programs	DR. RHYAN V. MOLINAR Campus Director	DR. ARMANDO A. TORRES Vice President for Campuses

THE COUNTRY'S 1st POLYTECHNICU

